Loot or Love? Will a Fortnite Player Find the Pirate Shop Token Before a Girlfriend?

D. Thompson*

Independent Researcher

November 2026 *Preprint, not peer reviewed*

Abstract

The Pirate Shop Token in Fortnite drops from 1 in 137,500 chests and makes everything in the Item Shop that costs V-Bucks free for 14 days. I wanted to know if we'd find one before we found a girlfriend, so I simulated both. The median solo player needs about 3,240 hours to find the token. Squads need more like 1,250 because when one person in a squad gets it, everyone does. For a guy who's looking and plays 10 hours a week, a girlfriend is roughly three times as likely as the token before the season ends, and that was true in all 400 versions of the model I ran. The token only pulls ahead past about 23 hours a week (12 in squads), and that's mostly because by then we've stopped leaving the house. Looks and personality matter a lot, but so does actually trying, and an average guy who makes an effort has around an 8.5% chance by Christmas. I also worked out that the survival time which gets you the most chests per hour is 5 minutes 49 seconds, so you can stop trying to win.

1 Introduction

Epic confirmed the token's drop rate in early November 2026: $p = 1/137{,}500$ per chest [1, 2]. Before that, people online were saying 1 in 700,000 [3]. So the real odds are five times better than everyone thought, which upgrades the token from impossible to just pointless.

Nobody thinks in chests, though, so I turned it into hours. Then I needed something to compare it to, and the obvious other thing people like us are failing to get is a girlfriend. In 2010 an economist called Peter Backus used the Drake equation (the one for counting alien civilisations) to work out why he was single [5]. I've stolen his method for the girlfriend side and added looks, personality, effort and the fact that every hour in Fortnite is an hour not spent meeting anyone.

2 Model

2.1 Token

Each simulated player's average survival time in match n is

$$\bar{T}_n = (1-\phi)^k \big[T^\infty + (T_0 - T^\infty)e^{-n/\tau}\big], \qquad (1)$$

starting at $T_0 = 14$ minutes and dropping towards about 6 once skill-based matchmaking (SBMM) works out that you're not as good as your first few games suggested. The factor $(1-\phi)^k$ is fatigue. I set it at 1% worse per game in a session, which is probably too kind, since in League of Legends win rates drop by more than 10% between the first and last game of a session [6].

Chests opened per match are Poisson with mean

$$\mathbb{E}[c_n] = \sigma C\big(1 - e^{-T_n/\kappa}\big), \qquad (2)$$

with $C = 12$ and $\kappa = 6$ minutes. It's an exponential because most chests get opened in the first few minutes after landing. σ is the share of chests you get yourself (1 in solos, 0.65 in squads). If anyone in your group finds the token, the whole group gets it [4], so in a squad all four players' chests count for you. On top of that each match costs about 4 minutes of lobby, queue and Battle Bus, and that grows as the queues get longer and we get slower at pressing ready.

2.2 Girlfriend

Backus treats every new person you meet as a long shot that has to pass a chain of filters. She's a woman (0.5), the right age (0.4), single (0.34 [7]), looking (0.35, the share of single women who are looking [7]), finds you attractive (1/20), you get on (1/10), you ask (0.5) and she says yes (0.5). That comes to about 1 in 33,600. I assume you meet about one new person a day, minus whatever socialising Fortnite has replaced [8]. (Guys our age average 0.51 hours a day socialising and 1.08 hours gaming, so gaming is already winning.)

DMs get the same treatment. I guessed that a cold DM gets a reply 8% of the time and that we send about three a week, which is probably generous. A reply then has to survive a conversation, a date and actually becoming a thing. Looks count for more online. On OkCupid the best-looking men got 11 times the messages of the worst-looking [11], so I used the same 11x gap for DMs.

Only half of single men are even looking for a relationship or dates, according to Pew [7]. So half the simulated guys are looking and half aren't, and the ones who aren't still occasionally fall into something, at a fifth of the rate. Every guy also gets a looks percentile and a personality factor r_j, drawn from a lognormal. r_j covers everything else, like confidence, being funny, or being able to hold a conversation without bringing up Fortnite. For Table 2 I added effort on top, at 1× for casually looking, 2.5× for actively trying and 5× for going all out. I made those numbers up.

About half of under-30s have used a dating app [9], so apps are in the model. Most people watching this on TikTok aren't on them, though, so the main results are for guys who are casually looking and don't use apps. Apps get a paragraph of their own in Section 3.

Since a lot of this is guessed, I tuned two unknowns until the model matched two real numbers: 63% of men aged 18–29 are single [7], and 39% of straight couples who got together in 2017 met online [12]. It turns out Backus's chain on its own is way too bleak. Taken literally, the average guy would wait 63 years. Once the model matches reality a guy who's looking waits around 2 years, so real life is something like 30 times kinder than meeting random strangers, probably because most people end up with someone they already know.

3 Results

3.1 Odds per try

Each new person you meet in person is about a 1 in 33,600 shot, and each cold DM about 1 in 3,500 (Figure 1). The DM figure leans on my guessed reply rate, so don't tattoo it anywhere, but even so a DM beats a chest by roughly 40 times. The only thing the token has going for it is volume. We open around 30 chests an hour and meet maybe one new person a day.

3.2 Hours to find the token

The median solo player needs 3,240 hours (Table 1), which is 135 days without sleeping. In a squad you open fewer chests yourself, but if anyone finds the token everyone gets it, so the median falls to 1,246 hours, 2.6 times faster than solo. Turns out you do need friends. Unfortunately a lot of you don't have any, so you're cooked. Region barely matters (3,200 hours in EU and NA, 3,500 where queues are longer). The thing I didn't expect was the lobby. Once SBMM has caught up with you, which takes about 800 games, matches get so short that around half (40 to 65%) of the time we spend "playing Fortnite" is actually staring at a loading screen.

Table 1: Hours of play to find the token. In duos and squads the token is shared, so the whole group's chests count.

Mode	Luckiest 10%	Median	Unluckiest 10%
Solo	480	3,244	10,800
Duo	279	1,902	6,339
Squad	180	1,246	4,136

3.3 Optimal time to die

Ignoring fatigue and randomness, chests per hour are $r(T) = 60C(1 - e^{-T/\kappa})/(T + O)$, where O is the time between matches.

Proposition 1. *$r(T)$ is maximised at $T^* = -\kappa\big[W_{-1}(-e^{-(1+O/\kappa)})+1\big] - O$, where W_{-1} is the lower branch of the Lambert W function.*

Proof. Setting $r'(T) = 0$ gives $e^{T/\kappa} = 1 + (T + O)/\kappa$. With $x = 1 + (T + O)/\kappa$ this becomes $-xe^{-x} = -e^{-(1+O/\kappa)}$. The principal branch gives $T < -O$, which is impossible, so $x = -W_{-1}(-e^{-(1+O/\kappa)})$, and this is a maximum. □

With $O = 4$ and $\kappa = 6$, $T^* = 5.82$ minutes, or 5 minutes 49 seconds, and the full simulation agrees (Figure 2). So once you've looted your immediate surroundings, it's actually better to leave the game. Mathematically there's no reason for your character to be alive, and let's be honest, you weren't going to win anyway.

3.4 Token vs girlfriend

The season ends on 5 December, 26 days after I started writing this. Figure 3 shows the chance of each by then for solo players who are casually looking and not on apps. Don't play at all and you've got about a 3% chance of a girlfriend and zero chance of the token. At 10 hours a week it's 2.5% against 0.8%. Every extra hour helps the token and hurts the girlfriend, and the lines cross at about 23 hours a week. The top 20% by looks can play up to 37 hours before the token catches them, and the bottom 20% only get 14. Over a lifetime at 10 hours a week the token comes first 30% of the time.

In a squad it's closer. Because the token's shared, a squad player at 10 hours a week has a 2.1% chance of it by December, and the crossover drops to about 12 hours.

3.5 Your odds

The averages hide a huge spread. Figure 4 is for a guy who's actively trying, plays 10 hours a week and isn't on apps, and it shows his chance of a girlfriend by Christmas depending on his looks and personality. Table 2 does the same for each level of effort. You can't

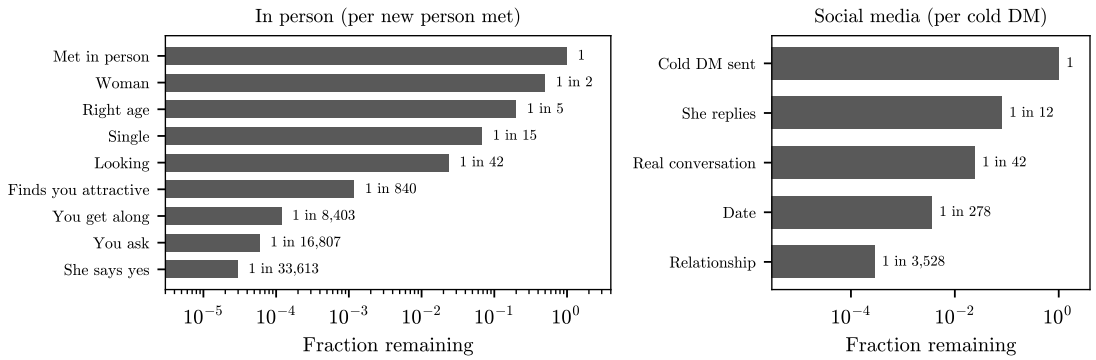


Figure 1: Chance of ending up in a relationship, per new person met (left) and per cold DM on social media (right). The token is 1 in 137,500 per chest.

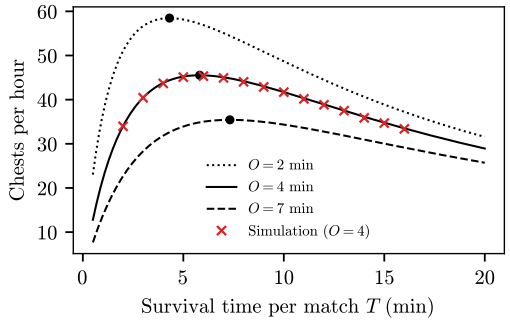


Figure 2: Chests per hour against survival time. Dots are T^*, crosses are the simulation.

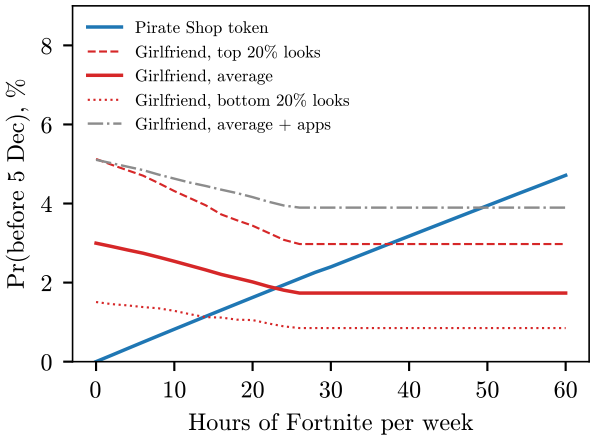


Figure 3: Chance of each before 5 December, by hours of Fortnite per week, for solo players who are looking and don't use dating apps. The grey line adds dating apps.

do much about your face before Christmas, but you can decide to actually try, and going from casually looking to actively trying more than doubles your odds.

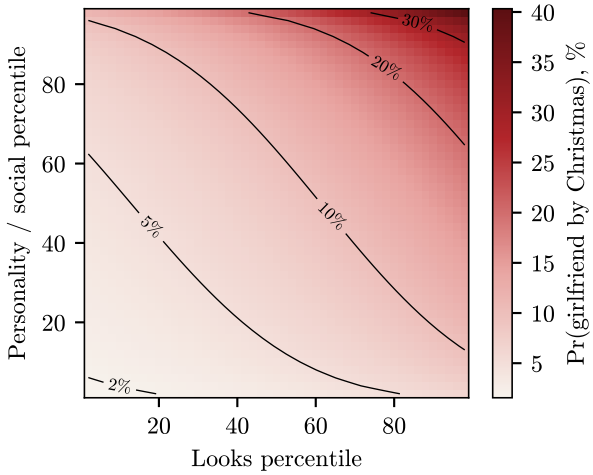


Figure 4: Chance of a girlfriend by Christmas, actively trying, 10 h/week of Fortnite, no apps.

Table 2: Chance of a girlfriend by Christmas (10 h/week, no apps). "Bottom" and "top" mean the 10th and 90th percentile for both looks and personality.

Effort	Bottom	Average	Top
Not looking	0.2%	0.7%	2.4%
Casually looking	1.0%	3.5%	11.7%
Actively trying	2.6%	8.5%	26.7%
Going all out	5.1%	16.3%	46.2%

3.6 Dating apps

A right swipe comes out at roughly 1 in 47,000, which is only about three times better than opening a chest. That's my estimate and not a real stat. The 0.6% match rate for men was measured [10], but every step after the match is guessed, so it could be anywhere from 1 in 14,000 to 1 in 100,000. What apps do help with is volume. Guys on them who are looking have nearly

double the chance by December (4.6% against 2.5% at 10 hours a week), and they can play up to about 49 hours a week before the token overtakes them. Rural guys run out of people to swipe on, so it helps them less.

3.7 How sure am I?

Not very, about the exact numbers, so I gave each of the 23 guessed parameters a range and reran everything 400 times with random values, recalibrating each time. At 10 hours a week the girlfriend beat the token in every one of the 400 runs (solo, looking, no apps). Where the lines cross is much less certain. It moved around from 17 to over 60 hours a week (90% interval), and in 1 run in 10 the token never caught up at all. According to the Sobol indices, the two guesses that matter most are how many chests we open per game and how often relationships end.

3.8 How to chase both

All that lobby time is wasted, so we might as well use it. At 10 hours a week the model has us playing about 48 matches, which is 48 trips back to the lobby. Send one DM from each, instead of three a week, and an average guy who's casually looking goes from a 3.5% chance of a girlfriend by Christmas to 9%. Add the 5:49 rule and leave once you've looted, and you fit in about 53 matches a week. The token goes from 0.8% to 1.1% by 5 December and the girlfriend to 9.6% by Christmas. Do it in a squad and the token's at 2.7%.

So the optimal strategy is to drop with a squad, loot, die on purpose at 5 minutes 49 seconds, and slide into someone's DMs while you wait for the bus. The model does assume the reply rate stays at 8%, which I doubt it would once you're sending 50 DMs a week from a loading screen.

4 Limitations

Epic doesn't publish chests per game or queue times, so I estimated them. The dating numbers are for young American men in general, and I'd guess Fortnite players do worse. Also, none of this shows that Fortnite makes you single. It could easily be that being single makes you play Fortnite.

5 Conclusion

So, token or girlfriend? For anyone actually looking, the girlfriend, and it isn't close. At 10 hours a week with no dating apps we have about a 2.5% chance of a girlfriend before 5 December and a 0.8% chance of the token. The girlfriend won in all 400 versions of the model.

People love quoting a "1% chance" of getting a girlfriend. It looks that bad because half of single guys aren't trying at all. An average guy who actively tries has about an 8.5% chance by Christmas, and going all out roughly doubles it. A guy in the top 10% for looks and personality who goes all out is at 46%. Meanwhile no amount of effort changes a drop rate. You need about 3,240 hours just to have a 50/50 shot at the token.

The token does win eventually, past about 23 hours a week (12 in squads), but only because by then we've stopped meeting people. For the bottom 20% by looks that happens at 14 hours, which is brutal.

If you still want the token, play squads, keep sessions short and leave once you've looted. That's still a median of 1,250 hours, or 52 days straight (solo it's 135). If you want both, send a DM from every lobby while you wait, which on its own takes an average guy from 3.5% to about 9% by Christmas.

A cold DM is about 1 in 3,500 and a chest is 1 in 137,500, so per try the DM is about 40 times better, and none of us are sending them. Apps roughly double your chances as well, and push the crossover out to 49 hours a week.

If you've actually found the token, comment how many hours it took. You'd be the first real data point in this paper.

References

[1] Fortnite Status (@FortniteStatus). Treasure drop rates. Post on X, November 2026.

[2] Video Games Chronicle. "Fortnite officially confirms Pirate Shop Access Token drop rate." 9 November 2026.

[3] FRVR. "Fortnite Pirate Shop token drop rate explained, and it isn't 1 in 700,000." November 2026.

[4] GamesRadar+. "The Fortnite pirate shop token explained." 5 November 2026.

[5] Backus, P. "Why I don't have a girlfriend: An application of the Drake Equation to love in the UK." University of Warwick, 2010.

[6] Sapienza, A. et al. "Individual performance in team-based online games." *Royal Society Open Science* 5(6):180329, 2018.

[7] Pew Research Center. "For Valentine's Day, 5 facts about single Americans." 2023.

[8] U.S. Bureau of Labor Statistics. *American Time Use Survey: 2025 Results*. 2026.

[9] Pew Research Center. "Key findings about online dating in the U.S." 2 February 2023.

[10] Tyson, G. et al. "A first look at user activity on Tinder." *ASONAM*, 2016. arXiv:1607.01952.

[11] Rudder, C. "Your looks and your inbox." OkCupid blog, 2009.

[12] Rosenfeld, M. J., Thomas, R. J. and Hausen, S. "Disintermediating your friends." *PNAS* 116(36):17753–17758, 2019.